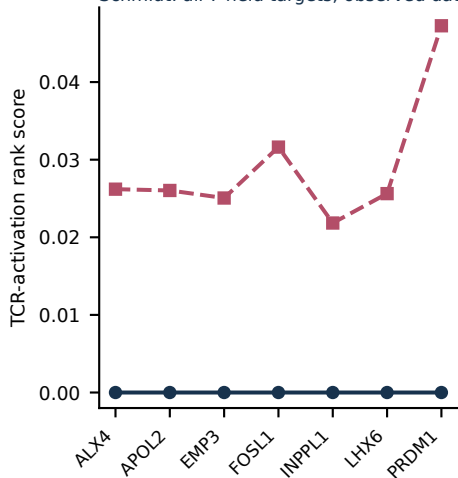


a Aggregation changes the target

Schmidt: all 7 held targets; observed data only



● Score of observed mean profile
 ■ Mean observed per-cell score

b OP3 program concordance

FP-ridge *	0.81	0.28 2/4	0.20 1/4
biolord	0.52	0.25 2/4	-0.20 2/4
CellFlow	0.65	NA	-0.29 1/4
CellOT	NA-P	NA	NA
PRnet	0.51	NA	-0.14 1/4
scPRAM	0.07 3/4	NA	-0.05 1/4
scFoundation †	NA-P	NA	NA
scGPT †	NA-P	NA	NA
	Type-I IFN	NF-κB	Effector (lymphocyte)

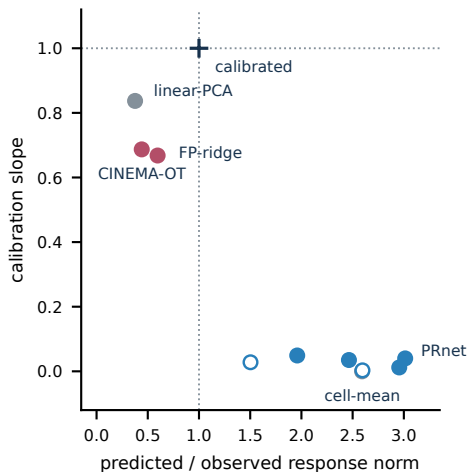
Mean over the lineages where the score is defined;
 n/4 printed where fewer than four.

NA-O: constant observed target; NA-P: constant prediction; NA: both, in different lineages.

Never plotted as zero.

† Adapted * Diagnostic comparator

c Direction is not magnitude



Six conditioned predictors overstate the response 1.5 to 3.0 times with slopes at or near zero; the three that understate it keep slopes well above zero. scPRAM predicts no shift.

d Checkpoint assay context

Frangieh protein fit; 124 held KOs

